Appendix

Supplementary Results

Experiments 1 is reported in full in the main paper and are not repeated here. This appendix supplements Experiments 2-4 with results on synthetic datasets at 10% and 20% missingness, complementing the 30% results reported in the main paper. The additional results show patterns similar to those at 30%, supporting the consistency of our findings across missingness rates.

A. Imputation Reshapes More Than Missing Values

We first report the structural-distortion results for the additional missingness rates, covering correlation structure, class separability, and marginal distributions.



Fig. 1: ΔCorr of all 46 imputation methods, grouped by category and sorted by median damage, with the same row order in all panels. The three left panels use synthetic MCAR, MAR, and MNAR missingness on eight datasets at a missing rate of 10%; the rightmost panel uses six real-world datasets. Dots are per-dataset medians on a log-scaled axis, and the dashed line marks the observed minimum.

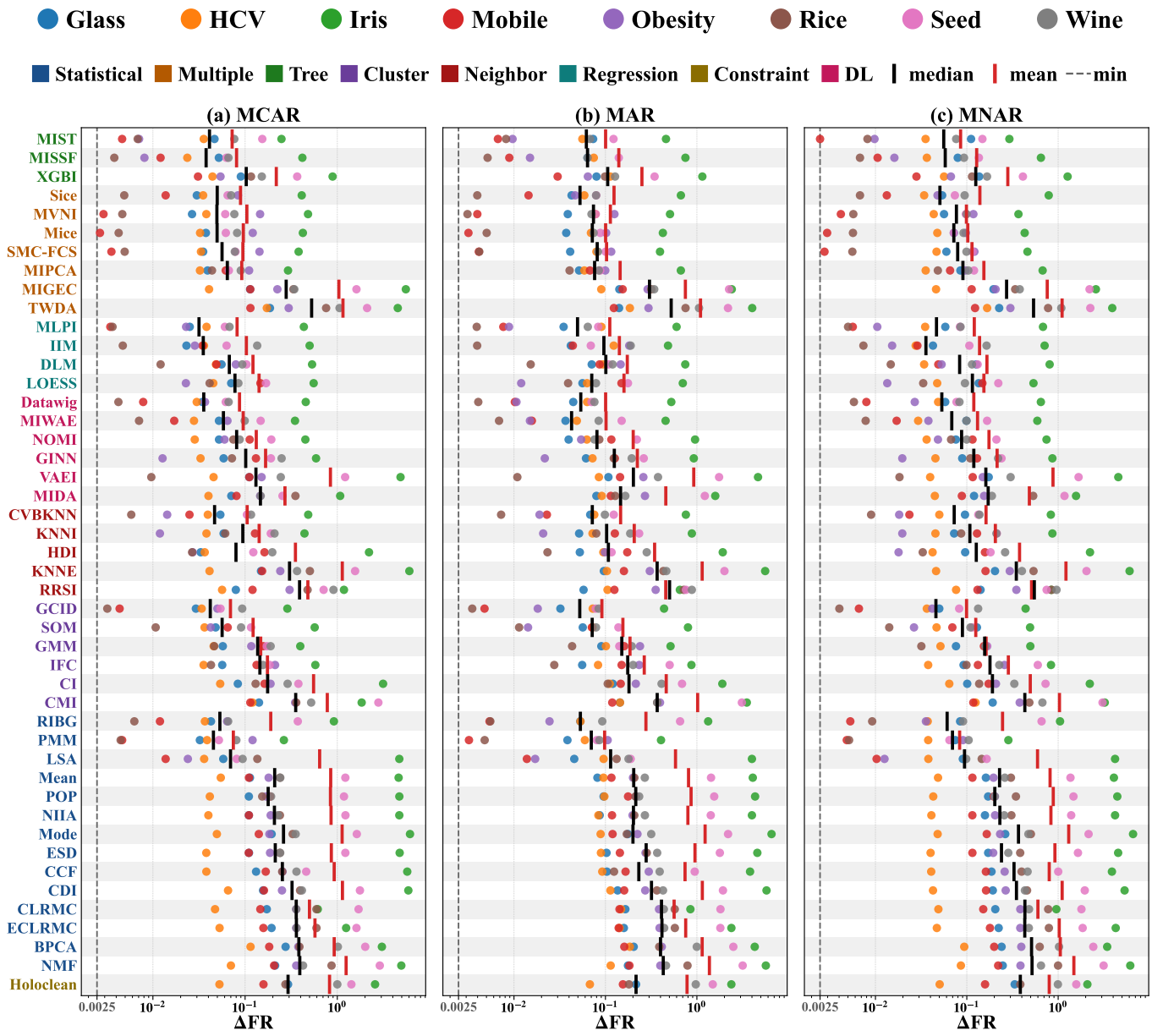


Fig. 2: ΔFR of all 46 imputation methods, grouped by category and sorted by median damage, with the same row order in all panels. The three left panels use synthetic MCAR, MAR, and MNAR missingness on eight datasets at a missing rate of 10%; the rightmost panel uses six real-world datasets. Dots are per-dataset medians on a log-scaled axis, and the dashed line marks the observed minimum.

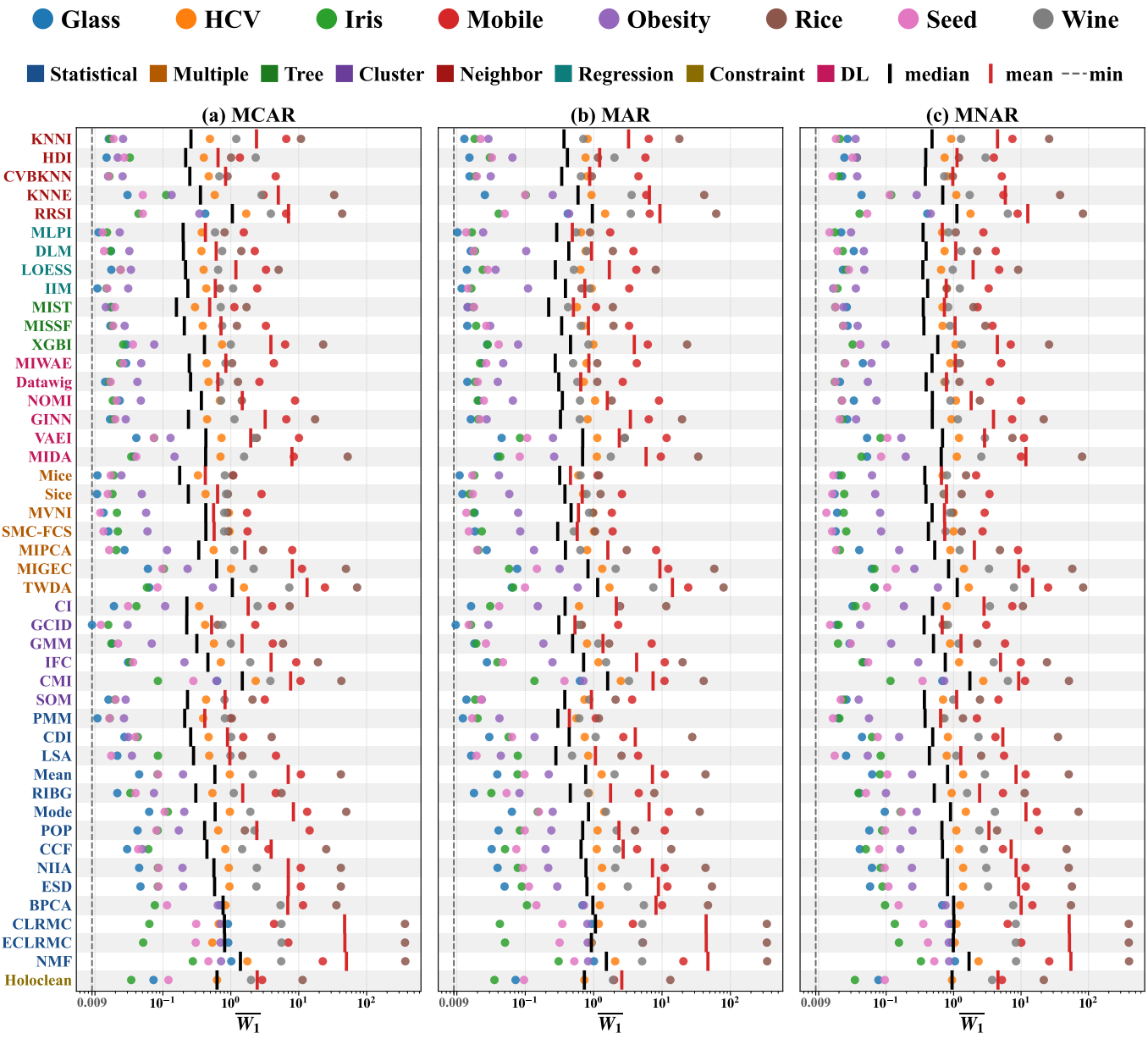


Fig. 3: $\overline{W_1}$ of all 46 imputation methods, grouped by category and sorted by median damage, with the same row order in all panels. The three left panels use synthetic MCAR, MAR, and MNAR missingness on eight datasets at a missing rate of 10%; the rightmost panel uses six real-world datasets. Dots are per-dataset medians on a log-scaled axis, and the dashed line marks the observed minimum.

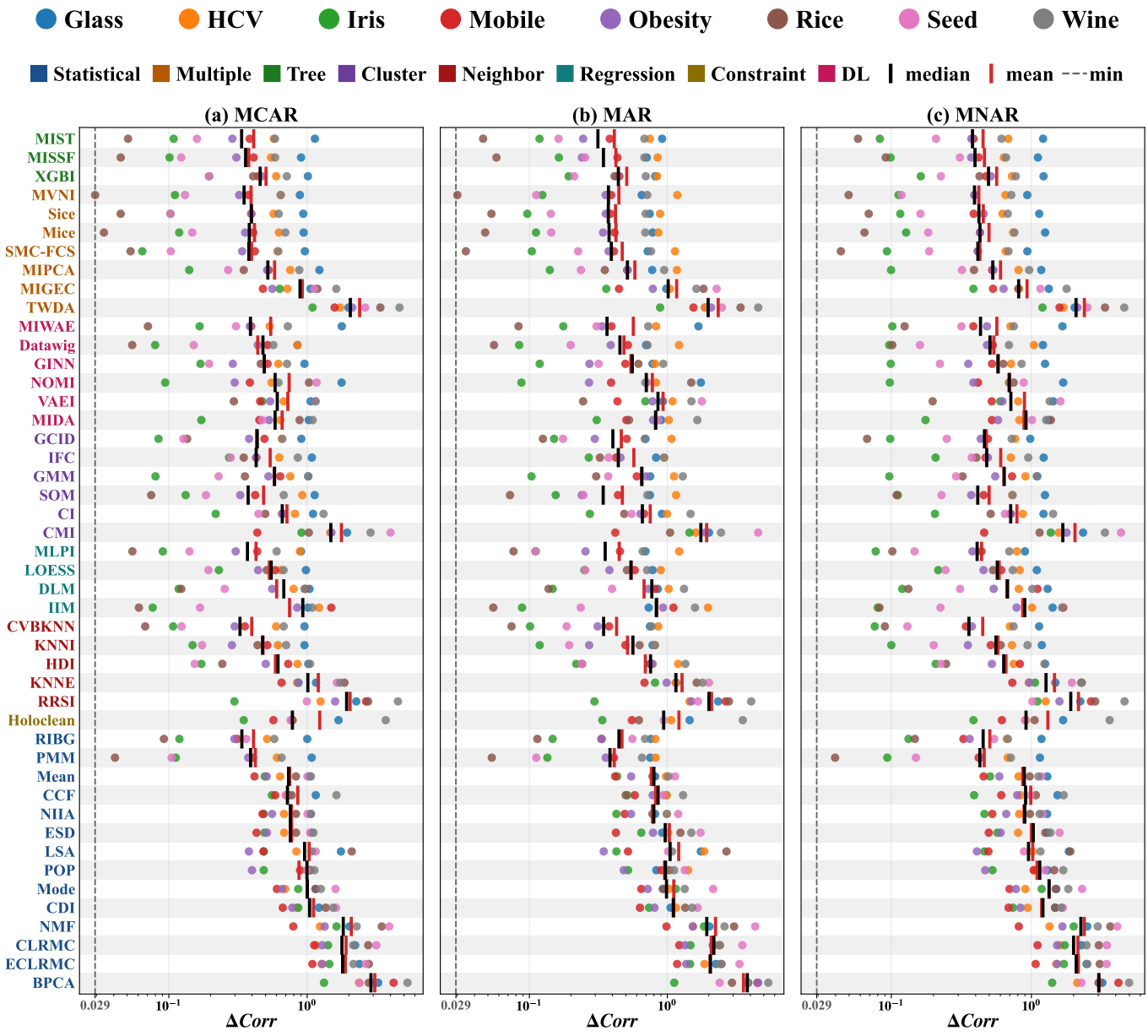


Fig. 4: ΔCorr of all 46 imputation methods, grouped by category and sorted by median damage, with the same row order in all panels. The three left panels use synthetic MCAR, MAR, and MNAR missingness on eight datasets at a missing rate of 20%; the rightmost panel uses six real-world datasets. Dots are per-dataset medians on a log-scaled axis, and the dashed line marks the observed minimum.

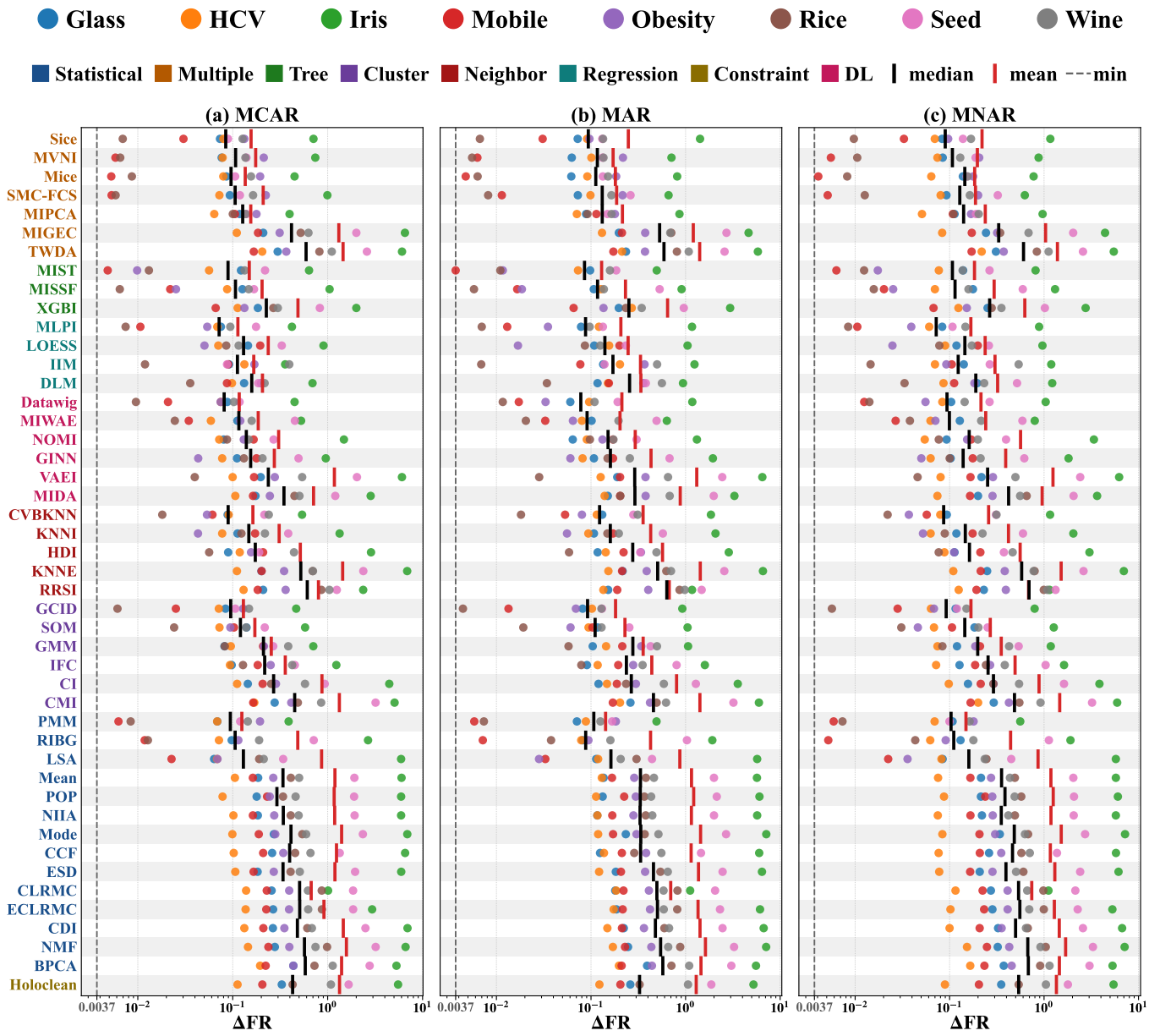


Fig. 5: ΔFR of all 46 imputation methods, grouped by category and sorted by median damage, with the same row order in all panels. The three left panels use synthetic MCAR, MAR, and MNAR missingness on eight datasets at a missing rate of 20%; the rightmost panel uses six real-world datasets. Dots are per-dataset medians on a log-scaled axis, and the dashed line marks the observed minimum.

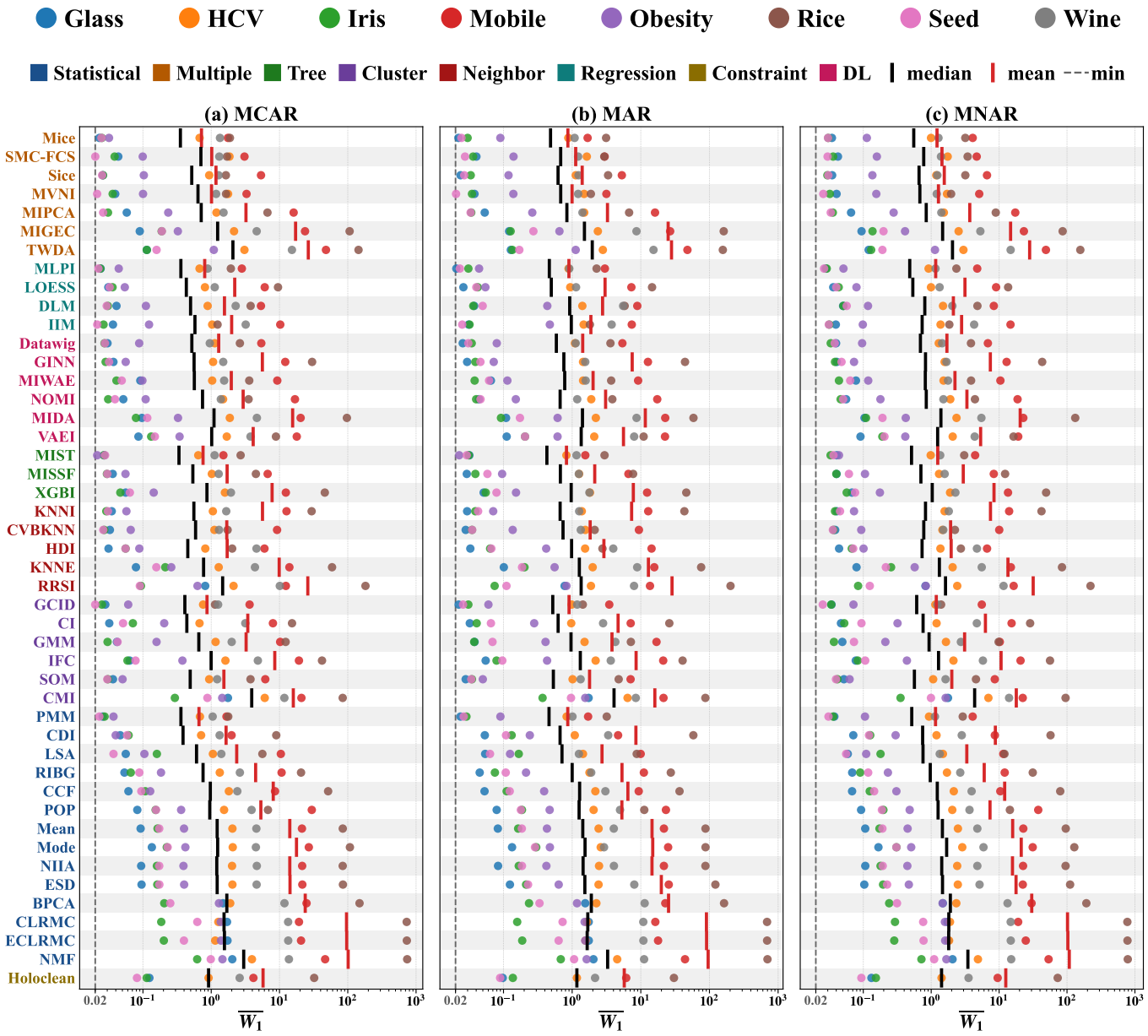


Fig. 6: $\overline{W_1}$ of all 46 imputation methods, grouped by category and sorted by median damage, with the same row order in all panels. The three left panels use synthetic MCAR, MAR, and MNAR missingness on eight datasets at a missing rate of 20%; the rightmost panel uses six real-world datasets. Dots are per-dataset medians on a log-scaled axis, and the dashed line marks the observed minimum.

B. Not All Structural Distortions Hurt Equally

We next present the fixed-effects analyses at 10% and 20% missingness to examine whether the structural metrics explain downstream classification harm beyond NRMSE across these settings.

TABLE I: Fixed-effects regression of Harm F1 on NRMSE and a structural indicator S for synthetic data at a 10% missing rate. Models absorb dataset, classifier, and imputer fixed effects.

S	Mech.	β	95% CI	$p(\beta)$	$\beta^{\mathrm{std,w}}$
ΔCorr	MCAR	+0.0179	[+0.0109, +0.0249]	5.0×10^{-7}	+10.6%
	MAR	+0.0170	[+0.0110, +0.0230]	3.5×10^{-8}	+10.3%
	MNAR	+0.0164	[+0.0092, +0.0236]	8.0×10^{-6}	+10.1%
$\overline{W}_1$	MCAR	-1.72×10^{-5}	$[-5.38\times10^{-5}, +1.94\times10^{-5}]$	0.357	−0.7%
	MAR	-2.47×10^{-5}	$[-7.83\times10^{-5}, +2.89\times10^{-5}]$	0.367	−0.9%
	MNAR	-1.92×10^{-5}	$[-8.07\times10^{-5}, +4.23\times10^{-5}]$	0.541	−0.8%
ΔFR	MCAR	+0.0033	$[-1.56\times10^{-4}, +0.0067]$	0.061	+3.4%
	MAR	+0.0051	[+0.0017, +0.0084]	0.003	+4.7%
	MNAR	+0.0023	$[-6.56\times10^{-4}, +0.0053]$	0.127	+2.2%

TABLE II: Fixed-effects regression of Harm F1 on NRMSE and a structural indicator S for synthetic data at a 20% missing rate. Models absorb dataset, classifier, and imputer fixed effects.

S	Mech.	β	95% CI	$p(\beta)$	$\beta^{\mathrm{std,w}}$
ΔCorr	MCAR	+0.0199	[+0.0106, +0.0292]	2.7×10^{-5}	+10.7%
	MAR	+0.0198	[+0.0125, +0.0271]	1.1×10^{-7}	+11.1%
	MNAR	+0.0155	[+0.0070, +0.0240]	3.3×10^{-4}	+8.7%
$\overline{W}_1$	MCAR	-1.77×10^{-5}	$[-4.57\times10^{-5}, +1.03\times10^{-5}]$	0.215	−1.2%
	MAR	-1.86×10^{-5}	$[-5.66\times10^{-5}, +1.94\times10^{-5}]$	0.336	−1.2%
	MNAR	-2.13×10^{-5}	$[-7.14\times10^{-5}, +2.89\times10^{-5}]$	0.406	−1.5%
ΔFR	MCAR	+0.0074	[+0.0040, +0.0108]	2.1×10^{-5}	+7.0%
	MAR	+0.0080	[+0.0031, +0.0128]	0.001	+7.2%
	MNAR	+0.0061	[+0.0020, +0.0102]	0.004	+5.5%

C. Hard Samples Reveal Classification Harm

We further extend Experiment 4 to synthetic datasets with 10% and 20% missingness. For each setting, we examine the panel-wise correlations along the chain from structural distortion (M_1) to hard-sample inconsistency (M_2 = HSIR) and downstream classification harm (Y = Harm F1). These additional results complement the 30% analysis in the main paper and assess whether the decision-layer relationships persist across missingness rates.

TABLE III: Pearson correlations for the $M_1 \to M_2 \to Y$ chain on synthetic datasets at 10% missingness, aggregated over 216 panels (dataset × classifier × mechanism). Within each panel, correlations are computed across imputers. The last two columns report the percentages of panels with $r > 0.30$ and $r > 0.50$, respectively. M_2 = HSIR and Y = Harm F1.

Path	Med. r	$[Q_1, Q_3]$	% > 30	% > 50
Shared decision-layer link				
HSIR → Harm F1 ($M_2 \to Y$)	0.843	[0.724, 0.939]	97.2	91.2
Panel A. $M_1 = \Delta$Corr				
M_1 → HSIR ($M_1 \to M_2$)	0.308	[0.166, 0.492]	51.4	23.6
M_1 → Harm F1 ($M_1 \to Y$)	0.189	[0.082, 0.315]	27.3	10.6
$\Pr(r_{M_2\to Y} > r_{M_1\to Y})$		98.1%		
Panel B. $M_1 = \Delta$FR				
M_1 → HSIR ($M_1 \to M_2$)	0.466	[0.327, 0.586]	80.1	42.1
M_1 → Harm F1 ($M_1 \to Y$)	0.370	[0.259, 0.500]	63.4	25.0
$\Pr(r_{M_2\to Y} > r_{M_1\to Y})$		95.8%		

TABLE IV: Pearson correlations for the $M_1 \to M_2 \to Y$ chain on synthetic datasets at 20% missingness, aggregated over 216 panels (dataset × classifier × mechanism). Within each panel, correlations are computed across imputers. The last two columns report the percentages of panels with $r > 0.30$ and $r > 0.50$, respectively. M_2 = HSIR and Y = Harm F1.

Path	Med. r	$[Q_1, Q_3]$	% > 30	% > 50
Shared decision-layer link				
HSIR → Harm F1 ($M_2 \to Y$)	0.740	[0.604, 0.836]	96.8	87.5
Panel A. $M_1 = \Delta$Corr				
M_1 → HSIR ($M_1 \to M_2$)	0.528	[0.342, 0.654]	79.6	53.2
M_1 → Harm F1 ($M_1 \to Y$)	0.350	[0.237, 0.473]	58.8	20.4
$\Pr(r_{M_2\to Y} > r_{M_1\to Y})$		89.8%		
Panel B. $M_1 = \Delta$FR				
M_1 → HSIR ($M_1 \to M_2$)	0.650	[0.524, 0.762]	93.5	76.9
M_1 → Harm F1 ($M_1 \to Y$)	0.563	[0.418, 0.732]	87.0	64.4
$\Pr(r_{M_2\to Y} > r_{M_1\to Y})$		79.2%		